

GRAVITATIONAL FIELD

1. Gravitational Field

What is meant gravitational field?

A region where a mass/body experiences a force of attraction due its mass.

The Earth has a **radial field** of gravity, which means that the gravitational field is circular and acts from the center point.

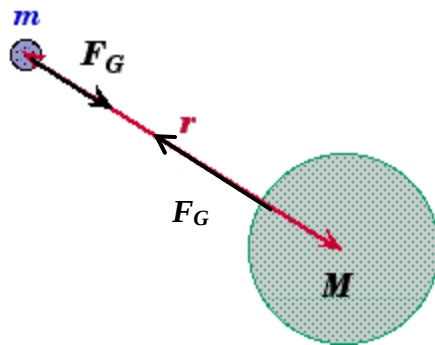
The Earths radial gravitational field is represented by the lines

Drawing the field lines ?



GRAVITATIONAL FORCE

The force of attraction is directly proportional to the product of masses (M_1 and M_2) of the bodies and inversely proportional to their separation (r) from the center of the bodies.



$$F \propto \frac{M_1 M_2}{r^2}$$

The equation for the gravitational force of attraction is given by:

$$F = G \frac{M_1 M_2}{r^2}$$

where, G is universal

gravitational
constant, $6.67 \times 10^{-7} \text{ Nkg}^{-2}\text{m}^2$

Base on N3L - both the bodies will experiences the magnitude of force but in opposite direction.

Ex: Apple and Earth

Assumption :

- i) the bodies are point masses.
- ii) their density is uniformly distributed.
- iii) Perfect sphere (if the bodies are rounded, ex. earth)

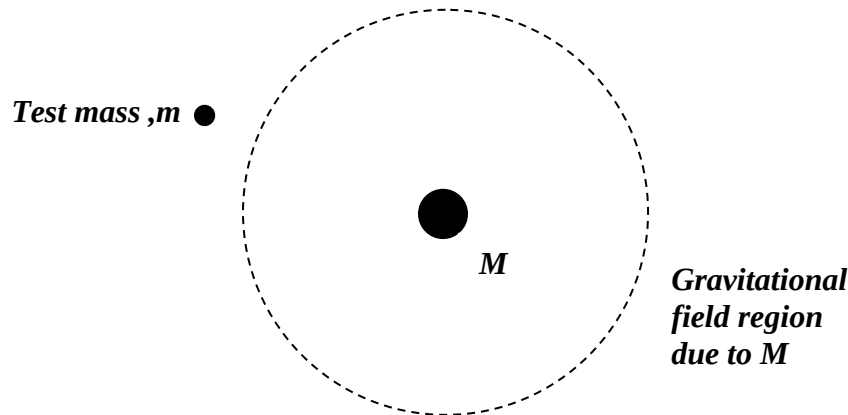
Ex:

1.

2. GRAVITATIONAL FIELD STRENGTH, g

Can be named as acceleration due to gravity and acceleration of free fall.

Gravitational field strength is defined as force acting per unit test mass when the test mass is placed in a region of gravitational field due to a larger mass.



Applying Newton's Second ($\mathbf{F}_{\text{net}} = \mathbf{ma}$) to the small object we can determine its acceleration if no other force are acting on the small mass $\mathbf{m}$ but gravity ($\mathbf{F}_{\text{net}} = \mathbf{F}_G$).

$$\begin{aligned}F_{\text{net}} &= F_G \\ma &= G \frac{mM}{r^2} \\a &= G \frac{M}{r^2}\end{aligned}$$

Because of its importance we usually give the acceleration of gravity due to the Earth a separate name,

$$\mathbf{g} = \frac{GM_E}{R_E^2}$$

* This equation **does not depend upon the mass of the small object** - the mass canceled out on both sides of the force equation. This proves that the acceleration due to gravity alone is the same on all objects independent of their mass, their size, their density, or even their speed.

* Local concentrations of mass in the Earth can change the magnitude of the acceleration of gravity in the third significant figure. In fact, it is often true that as you go up a mountain the value of $\mathbf{g}$ gets larger not smaller as you would expect because mountains contain a larger concentration of denser rocks.

* Since the value of $\mathbf{g}$ is not a fixed constant, in this course we will normally use 9.81 m/s^2 when solving problems.

Factors affecting g , the acceleration due to gravity:

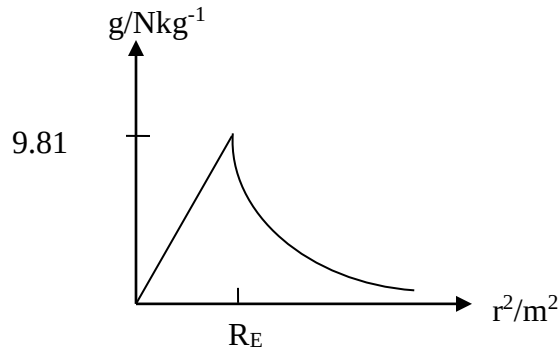
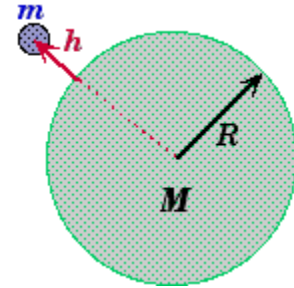
- i. altitude and depth
- ii. latitude
- iii. rotation of the earth.

Altitude and depth

If $h \ll R$, then

$$F_G \approx G \frac{mM}{R^2}$$

$$\begin{aligned} F_G &= G \frac{m_1 m_2}{r^2} \\ &= G \frac{mM}{(R+h)^2} \end{aligned}$$



Latitude

As we know that earth is not a perfect sphere, the radius at pole is slightly smaller than at equator, the gravitational field strength at equator is less than at pole.

Rotation of earth

What is the centripetal acceleration of a person standing on our planet at latitude 30 deg, caused by the rotation of the Earth? Earth's radius $R=6371\text{km}$. This is a mean value of the Earth's radius, which is accurate enough for the purpose of our problem.

Given:

$R = 6371\text{km}$ – mean radius of Earth at Equator

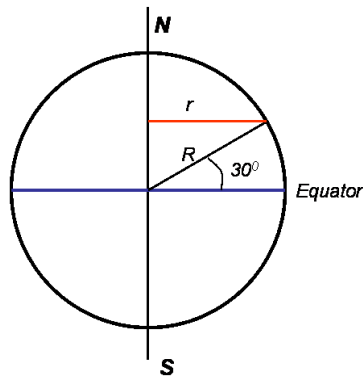
$T = 24\text{ h}$ – period of motion, we assume it is known to us.

Looking for: $a = ?$ – centripetal acceleration at altitude 30 deg.

To calculate this value we must find the radius of circular motion for the point on Earth at latitude 30 deg, we name it r .

From the formulas describing circular motion we have

$$a_c = \frac{v^2}{r} = \frac{\left(\frac{2\pi r}{T}\right)^2}{r} = \frac{4\pi^2 r}{T^2} \quad (1)$$



From trigonometry we find see

$$r = R \cos 30 \quad (2)$$

Substituting the numbers into formula (2) we get

$$r = 5517.5 \text{ km} = 5517500 \text{ m}$$

Substituting this value and value for period T into formula (1) we get

$$a_c = \frac{4\pi^2 r}{T^2} = \frac{2.18 \times 10^8}{7.47 \times 10^9} = 0.03 \text{ ms}^{-2}$$

This centripetal acceleration due to Earth's rotation is about 327 times smaller than gravitational acceleration g .

Therefore, the actual gravitational acceleration at this point is given by:

$$9.83 - 0.03 = 9.80 \text{ ms}^{-2}$$

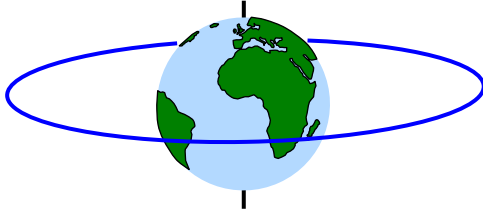
The weight of a mass 2 kg at this point is $2 \times 9.80 = 19.6 \text{ N}$

Ex : If the radius of Earth at equator is 6371 km and at pole is 6350 km, determine the weight of a man who has a mass of 60 kg.

3. ORBITS

An orbit is the path taken by one celestial body around another. We distinguish the bodies from each other by the terms orbiting body and central body.

Ex – earth and moon



Let's consider that the radius of Earth is R_E and the radius of orbit is R_O .

Remember that R_O is measured from the center of the Earth.

As the body rotates with uniform speed and radius,

(Required force), $F_C = F_G$ (Provision)

$$\frac{mv^2}{R_O} = \frac{GM_E m}{R_O^2}$$

$$v = \sqrt{\frac{GM_E}{R_O}}$$

Since $v = \frac{2\pi R_O}{T}$

$$\frac{4\pi^2 R_O^2}{T^2} = \frac{GM_E}{R_O}$$

$$R_O^3 = \frac{GM_E}{4\pi^2} T^2$$

**Equation that can be used to calculate the mass of central body.

Ex: Earth is orbiting around the Sun, if the distance from center to center of the bodies is 1.51×10^{11} m and the period is 365 days 5 hrs, calculate the mass of the Sun?

4. GEOSTATIONARY SATELLITES and POLAR SATELLITES

Geostationary satellites

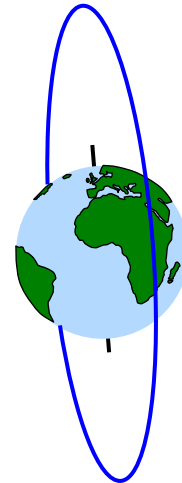
Conditions – i) orbits from west to east.
ii) period of 24 hrs.
iii) always above the equator line of earth.
iv) same angular speed as earth.

Applications – i) weather monitoring
ii) Telecommunication

Polar satellites

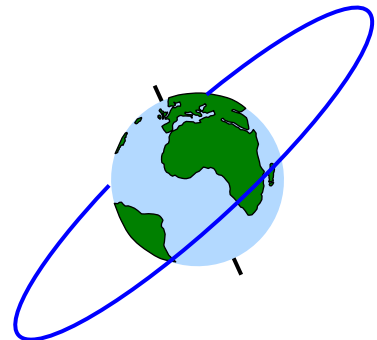
Conditions – i) rotates from pole to pole
ii) period of about 90 minutes to 105 minutes

Applications – i) weather forecasting
ii) world mapping
iii) spying



Possible orbits : All the orbits of the bodies **must** coincide with the center of the Earth.

What happens if the center of the orbit does not coincide?



5. GRAVITATIONAL POTENTIAL

Every point in a gravitational field has a scalar quantity called gravitational potential associated with it.

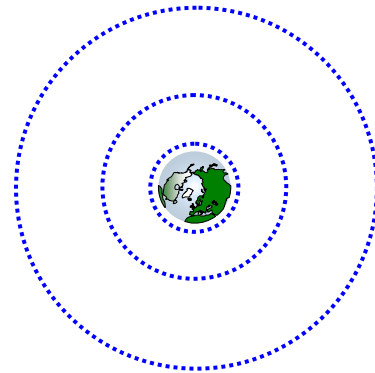
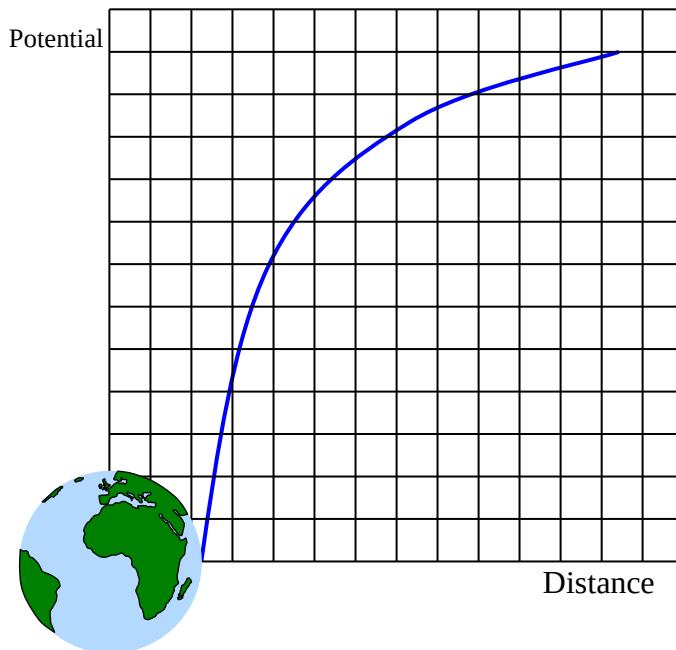
Definition: The gravitational potential, Φ at a point is the work done per unit test mass, m_t when the test mass is transferred from infinity to that point.

$$\phi = \frac{\infty \xrightarrow{\text{Workdone}} \text{point}}{m_t} \quad \text{unit : Jkg}^{-1}$$

Since work done is the product of force and the displacement moved by the body,
It can be written as that the gravitational potential at a point in the field:

$$\phi = -\frac{GM_E}{R_O}$$

Gravitational potential is a scalar quantity.



Equipotentials line
Equipotentials are always
perpendicular to the field lines.
Can you draw the field lines?

****** At infinity there is no gravitational force of attraction, thus as the body moves away from the center body, the gravitational potential became less. Therefore, the gravitational potential at any point in the field is given a negative value.

Remember [When you do work on a body to move it body to a higher place, the body gains gravitational potential. However, in actual situations, the gravitational potential becoming less negative and eventually approaches zero (a situation where no gravitational force acting on the body)]

Ex : The Earth has a mass of 5.98×10^{24} kg and a radius of 6370 km. Calculate the
i) gravitational potential at points A and B distance of 1500 km and 2500 km from Earth's surface.

ii) What is the change in the gravitational potential?

iii) How does the quantity in (ii) is conserved?

iv) Calculate the final speed of a body with a mass of 250 kg at A, when it is released from rest at B. Ignore the air resistance.